MODULE 7: RESEARCH WRITING
(CO: 7)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BENG101L Official Syllabus, VIT Department Question Bank, and VIT FAT Past Papers

Scope of This Textbook (All Syllabus Topics): (1) Interpreting and analyzing peer-reviewed research articles, (2) Approaches to Literature Review and Review Paper Writing (Narrative, Systematic, Meta-analysis), (3) Canonical structure of a technical research paper (IMRaD: Introduction, Methods, Results, and Discussion), (4) Citation and referencing standards: IEEE and APA style conventions, and (5) Academic integrity, ethics, plagiarism prevention, and peer review mechanics.

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1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 7

In VIT Final Assessment Tests, Module 7 carries 15 to 20 marks:

1. **Question Type A (10 Marks):** IMRaD Research Paper Structure: Explaining the detailed purpose, content, and writing conventions of Introduction, Methodology, Results, and Discussion sections in engineering journals.
2. **Question Type B (10 Marks):** Referencing & Academic Ethics: Formatting 5 diverse sources (journal article, conference paper, patent, book, online dataset) according to IEEE citation style and outlining ethical protocols to prevent plagiarism.

Research Element	IMRaD Section	Key Scientific Requirement
Introduction	Broad context -> Literature gap -> Research question / hypothesis	The inverted pyramid: moves from general background to specific objective
Methodology	Hardware setup, datasets, algorithms, simulation parameters, repeatability	Must provide sufficient detail for an independent researcher to replicate
Results	Empirical findings, figures, tables, performance graphs	Pure objective data presentation without speculative interpretation
Discussion	Data interpretation -> Comparison with literature -> Limitations -> Future work	Explains the physical or algorithmic meaning behind observed findings
IEEE Referencing	Numeric bracketed citations: [1], ordered sequentially as they appear in text	Standard across electrical, electronics, and computer science fields

2 Topic 1: The IMRaD Structural Framework

2.0.1 Anatomy of an Engineering Research Paper

1. **Title & Abstract:** Concise, descriptive title with high-density keywords. Structured 200-word abstract summarizing background, method, principal findings, and significance.
2. **Introduction:**
   - Contextualizes the engineering problem.
   - Reviews existing literature to establish current state of the art.
   - Identifies the research gap that motivated the study.
   - Formulates the exact research hypothesis and study scope.
3. **Methodology (Materials & Methods):**
   - Complete description of materials, component specifications, datasets, and benchmark baselines.
   - Mathematical equations and pseudo-code for algorithms.
   - Experimental procedures and statistical error bounds.
4. **Results:**
   - High-resolution data plots with error bars.
   - Comparative tables benchmarking performance against previous literature.
5. **Discussion:**
   - Explains the mechanisms driving the observed results.
   - Addresses discrepancies or unexpected anomalies.
   - Outlines study limitations and promising directions for future inquiry.

3 Topic 2: Referencing Standards (IEEE vs APA) & Academic Integrity

3.0.1 IEEE Citation Format Conventions

IEEE style uses bracketed numbers in order of appearance in the text:

- **Journal Paper:** [1] J. K. Author and R. M. Co-Author, "Title of journal article," **IEEE Trans. Power Syst.**, vol. 35, no. 4, pp. 2890-2898, Jul. 2020.
- **Conference Paper:** [2] P. S. Engineer, "Deep learning models for fault detection," in **Proc. IEEE Int. Conf. Autom.**, Chennai, India, 2024, pp. 112-117.
- **Book:** [3] C. L. Smith, **Advanced Digital Signal Processing**, 3rd ed. New York, NY, USA: Wiley, 2022.

3.0.2 Academic Integrity & Plagiarism Prevention

- **Direct Quotes:** Enclose in quotation marks with exact page citation.
- **Paraphrasing:** Rewrite in your own original syntax and vocabulary, and cite source.
- **Self-Plagiarism:** Re-using your own previously published text without citation is a serious academic infraction under IEEE and UGC ethical codes.

4 Quick Self-Test Checklist Before Your Exam

1. Diagram the four components of the IMRaD research paper architecture.
2. Format a journal paper, conference proceeding, and textbook using IEEE reference style.
3. Differentiate between Results (objective data) and Discussion (interpretation).
4. Explain the difference between mosaic plagiarism and proper academic paraphrasing.